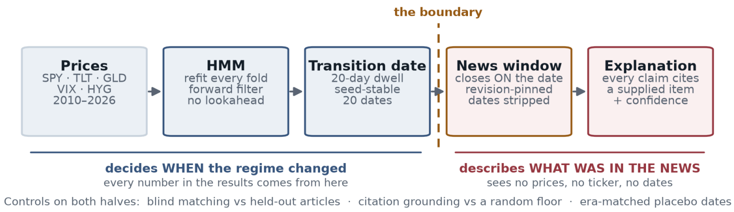
Regime detection that can say what happened — and a measurement of how much that is worth

An auditable explanation layer for any regime model. BSc Mathematics with Statistics, University of Bristol.

Finding: the explanations are specific and faithful to their sources; whether they are *diagnostic* of a regime change is unresolved, and this design could never have resolved it.



PROBLEM STATEMENT

1 · No explanation. A regime model tells you the market changed, not what happened — it only sees returns. A human fills the gap from memory, so one signal becomes thirty private readings that leave when the people do.

2 · No verification. The obvious fix makes it worse. A language model answers in seconds, and it reads the same whether drawn from that month's news or invented. An unverifiable explanation beats no explanation only in the way that feeling informed beats being informed.

SOLUTION OVERVIEW

The HMM decides *when* the regime changed; the language model only describes *what was in the news* then. No statistic comes from the model. For each date it retrieves a 14-day news window and returns an explanation whose every claim cites a supplied article.

The window **provably** closes on the date: a window object refuses to construct if any item post-dates it. Pages are pinned to their revision id, because a mean of **38.2%** of a Wikipedia day-page as it stands today (peak 71.2%) was written *after* the day it describes.

USE OF AI

`claude-opus-5` via the Anthropic Messages API — one call per window, effort high, schema enforced server-side. Prompts are version-controlled files with a dated iteration history, never inline strings; every call logs the model id, prompt hash and input hash, so any sentence traces to the exact prompt and news behind it. Built with Claude Code. MediaWiki API, yfinance, hmmlearn, scikit-learn, Streamlit.

VALIDATION

Can you trust the explanations?	What we found
Date hidden: can we still tell which two weeks it describes?	72% — guessing gives 33%
Claims really come from the source cited	96.4% — a random source gives 1.8%
Invented citations	none, in 474 claims
Confident on real events, vs ordinary days	**80% vs 62% — ignoring the news gives no gap**
Whole study re-run on a different AI	same scores

The detector underneath: stressed days move **2.18×** as much as calm days out-of-sample, and 18 of 20 folds separate independently ($p = 0.00040$).

IMPACT & VALUE

Who needs it. A risk team asked why a stress signal fired that week. A PM handing a book to someone who was not there. A researcher with a changepoint model and no way to say what its dates mean. All three rely on memory today.

What changes. The interpretation becomes a written record with dated sources, produced the same way every time, and carrying a number for how far to trust it. **The reusable part is the controls, not the explanations** — point them at any regime model's dates, any asset, any method, and every explanation returns with the scorecard above attached.

Honest scope. Row four says these are not yet shown to tell real events from ordinary days — so this buys an auditable record of *why* a date was flagged, not a trading signal.

REFLECTIONS

The placebo arm went against me and is the result I would keep: power at that gap is only 0.17, and the constraint is 20 transitions, not controls, so more placebos could never have helped. **Next:** pre-specify volatility *onsets*, which roughly triples it. One leak stays open — the boundary is 23:59 UTC and Wikipedia reports the US close that evening.